

# A Scenario-Knowledge-Driven Pipeline for Just-in-Time Assistance

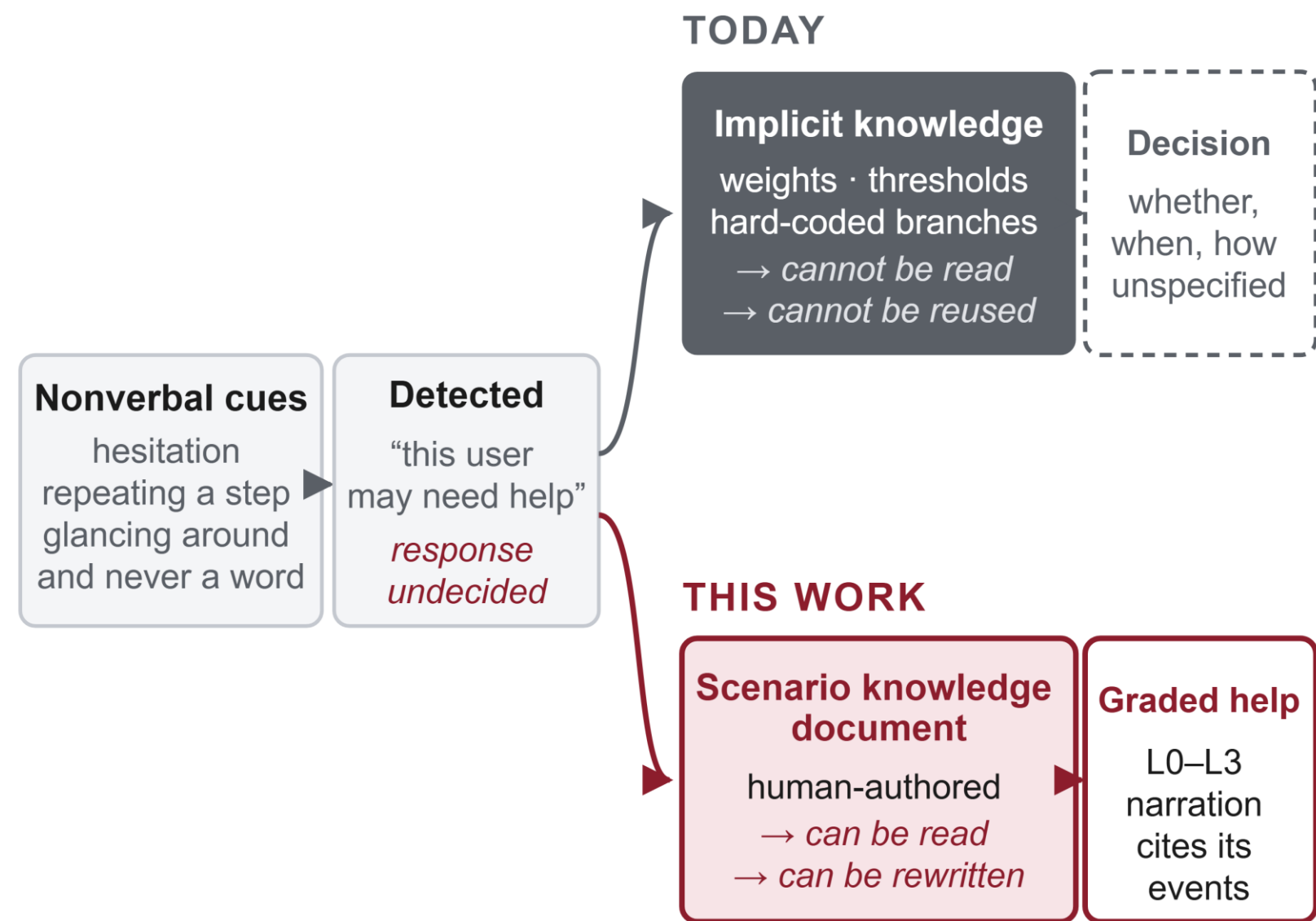
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## 1 FROM DETECTION TO DECISION

Silent strugglers at public kiosks hesitate, repeat a step, glance around, and never ask. Our earlier work showed that a machine can read that signal before the user says anything [1]. A detection, however, does not decide the response. Just-in-time assistance [2] asks whether, when, and how to help. That knowledge is scenario-specific. Today it sits in code, where nobody can read or reuse it.



The same detection, answered either by implicit knowledge or by an explicit document.

## 2 KNOWLEDGE AS A DOCUMENT

One document holds what the kiosk scenario assumes, and three modules read it.

- The rule layer (M1) emits only the events named in its signals section.
- The LLM narrator (M2) reads the signals section.
- The LLM assessor (M3) reads the whole document.

A new scenario means a rewritten document and retuned rule thresholds. The modules stay.

**kiosk\_v1 · scenario knowledge document**  
fixed on 2026-07-07, before any case run  
written from the interface design, the study protocol, and two sessions other than Case A and B

**Scenario**  
someone books a travel ticket at a public touch-screen kiosk; an attendant is near but must not hover; struggle precedes asking

**1 Task structure**  
easy warm-up (~2 min, linear) → hard variant with built-in friction: split seat screens · gated ticket-amount step · loading screens → loading ↔ passenger-info loop; payment only via ticket amount

**2 Observable signals (nine event types)** → read by M1 and M2  
screen change · page revisit · stall · long inactivity · tap burst · scroll thrash · expression spike · gaze aversion · posture shift  
evidence, not proof; read against the current screen and recency

**3 Plausible assistance-need states**  
hypotheses about the moment, not labels on a person: smooth\_progress · reading\_or\_deciding · silent\_struggle · active\_help\_seeking · recovering  
→ M3 reports a need level: none · monitor · likely · urgent

**4 Intervention options**  
L0 observe · L1 passive cue · L2 light inquiry · L3 contextual hint  
least intrusive first; skipping a rung needs strong evidence

**5 Social norms**  
public space · prefer under-intervention; never reveal the watching  
comparing, reading, or leaving the flow is reasonable, not an error

the whole page → the assessor (M3), in full, at every call

## 3 SIGNALS TO CITED NARRATION

**1 Nonverbal streams** features, not raw video · sensor rate

- scene · timeline · screen, revisit flag
- browser\_events · idle, taps, scroll
- face\_hd · 17 action units, 30 Hz
- gaze · head yaw
- body · torso lean

**evaluation data stays out**  
annotations, retrospective memos, identities: never in M1–M3 or in any prompt

**2 M1 rule layer · nine event types** fixed thresholds; the rules learn nothing

a screen segment starts → screen change  
segment flagged as revisit → page revisit  
no interaction ≥ 8 s → stall  
no interaction ≥ 20 s → long inactivity  
≥ 4 presses in 2 s → tap burst (click, burst)  
≥ 6 scroll events in 5 s → scroll thrash  
mean-AU variability,  $z > 2$  → expression spike  
yaw off median > P95, ≥ 1 s → gaze aversion  
torso lean rate > P97, 0.5 s → posture shift  
no repeat of a type within 5 s · source and rule kept

**one event, as omitted (Case A)**  
{ id: E12, t: 162.2 s, type: click\_burst, summary: "rapid repeated tapping (4 presses in 2 s)", provenance: { source: browser\_events, rule: press\_count\_window, window: [162.2, 163.2], value: 4 } }

**3 M2 narrator** event-triggered · append-only

events within 5 s → one batch  
prompt = narration so far  
\* new events as JSON  
output = new sentences only, each citing its events [E##]  
an uncited sentence → retry once, else dropped

At 02:42 the user produced another click burst of four presses in two seconds while still on the revisited H load2 screen [E12].  
Case A narration, verbatim

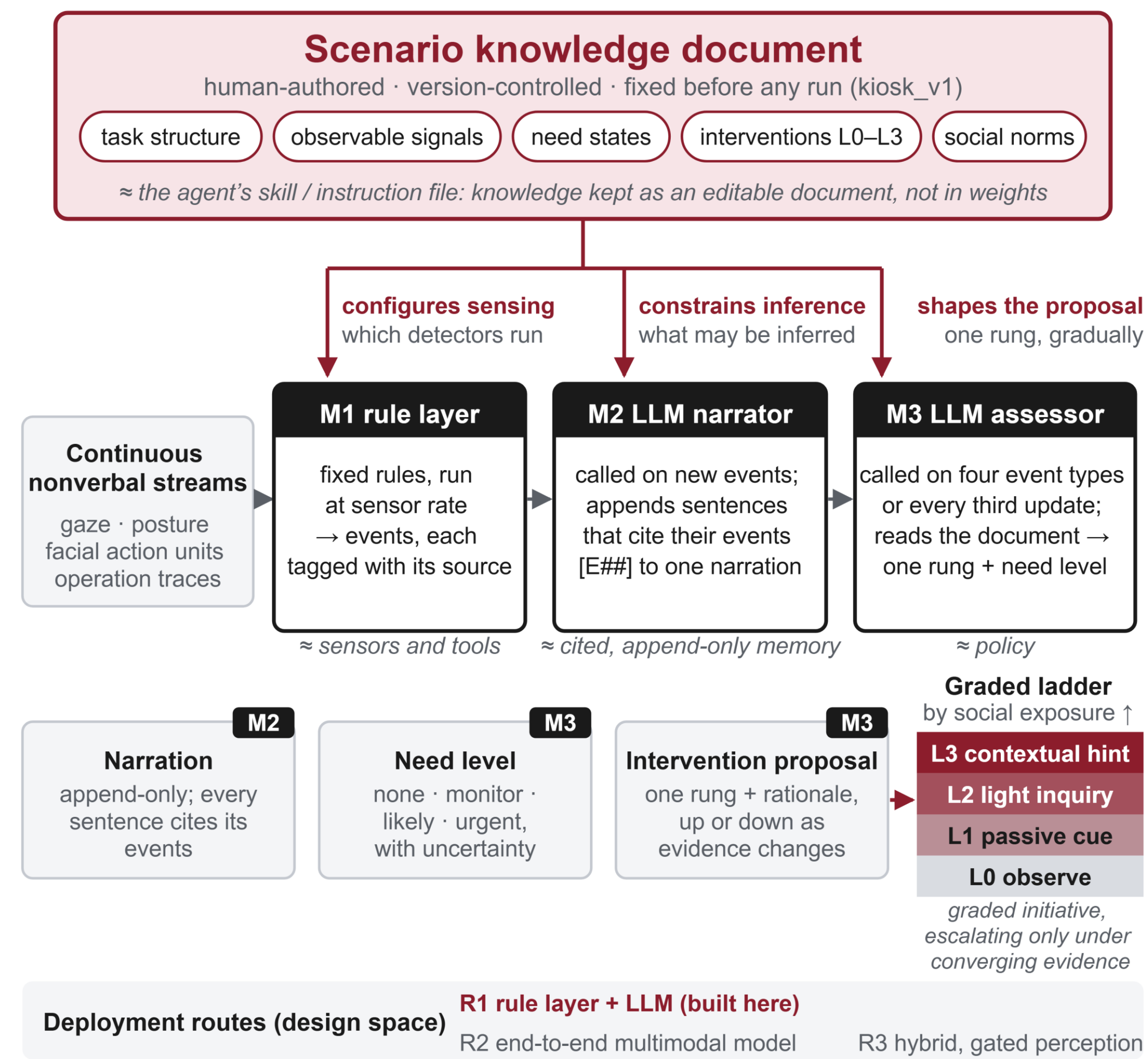
One real event, E12, from rule to record to the sentence that cites it.

## Writing down whether, when, and how to help as the scenario knowledge.

Detecting a kiosk user who struggles without asking is only the first step. One human-authored document carries the rest: what the sensing watches, what the LLM may infer, and how far a proposal may go, from silent observation to a hint. For a new scenario, the document changes and the pipeline stays.

## 4 FRAMEWORK

- The rule layer (M1) turns nonverbal streams into events.
- Events, not frames, trigger the LLM narrator (M2), which adds cited sentences to a running narration.
- Four event types (stall, long inactivity, tap burst, expression spike), or every third narration update, trigger the LLM assessor (M3). It reads the narration against the document and proposes one rung, L0 to L3, with a need level, none to urgent.



Rungs rise with social exposure: how much the intervention reveals that the user was watched.

### The pipeline in agent-framework terms

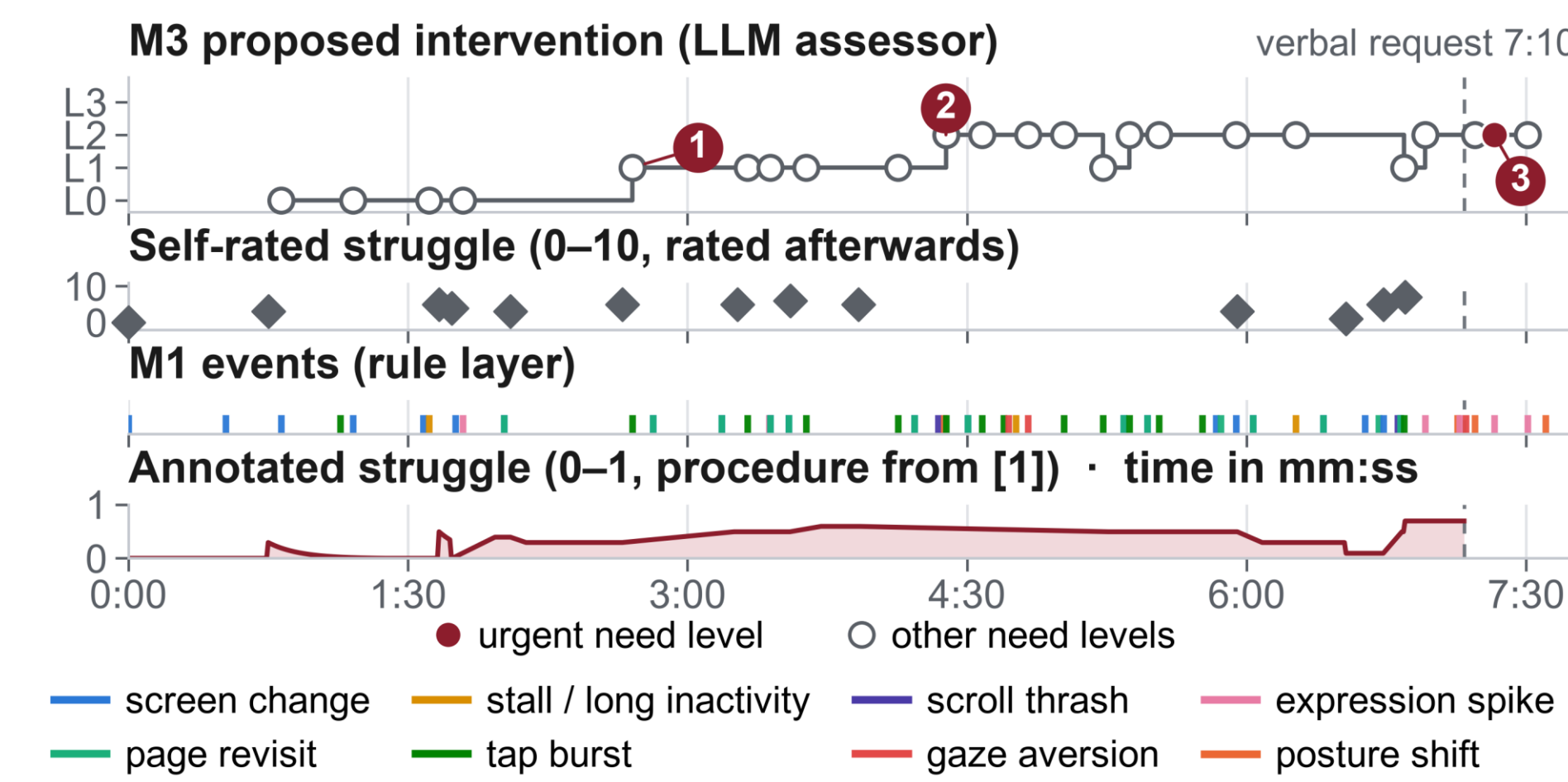
- the document ≈ a skill or instruction file · M1 ≈ sensors and tools · M2's narration ≈ cited, append-only memory · M3 ≈ policy
- event-triggered calls ≈ an event-driven loop, not per-frame captions

## 7 PROOF OF CONCEPT

### Case A, the dense session (430 s)

- 02:42 a tap burst at an already-revisited screen lifts L0 to L1
- 04:23 the L2 plateau begins, ≈3 min before the verbal request
- 07:20 the one urgent need level, after the request; the proposal stays at L2

Assessor, 07:20: "(L3) would over-intervene given remaining uncertainty ..."



430 s to the verbal request (dashed line). L2 held from 04:23, with two brief dips to L1, until the request.

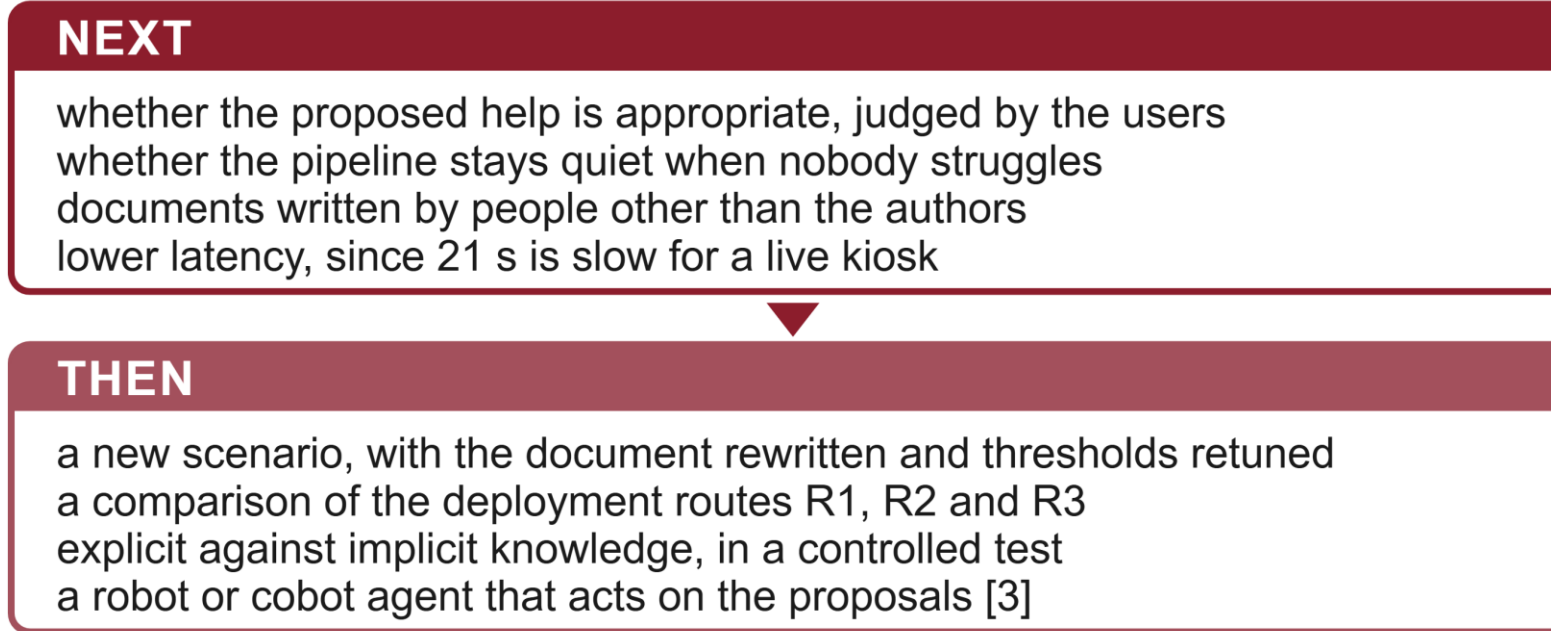
Per-case summary. Rungs: L0 observe, L1 passive cue, L2 light inquiry, L3 contextual hint.

|                                       | Case A (dense) | Case B (long) |
|---------------------------------------|----------------|---------------|
| Time to verbal request                | 430 s          | 865 s         |
| Struggle episodes detected (M1)       | 12 of 13       | 7 of 7        |
| Narration updates (uncited sentences) | 33 (0)         | 62 (0)        |
| Proposals L0/L1/L2/L3                 | 4/7/12/0       | 3/13/16/1     |
| Median assessor latency               | 20.8 s         | 22.4 s        |

Scope: two sessions, one model, one set of thresholds, one annotator.

## 8 RESEARCH AGENDA

Two cases show that the pipeline runs end to end. What they cannot show orders the work below.



## REFERENCES

- Z. Li et al., "Multimodal modeling of help-seeking intentions in self-service kiosk interactions toward just-in-time assistance," *Procedia CIRP*, vol. 140, pp. 1159–1164, 2026.
- I. Nahum-Shani et al., "Just-in-time adaptive interventions (JITAs) in mobile health: Key components and design principles for ongoing health behavior support," *Ann. Behav. Med.*, vol. 52, no. 6, pp. 446–462, 2018.
- D. Tanneberg et al., "To help or not to help: LLM-based attentive support for human-robot group interactions," in *Proc. IEEE/RSJ IROS*, 2024, pp. 9130–9137.

## 5 THE TWO SESSIONS

Two sessions from our kiosk study [1] are the cases, Case A and Case B, chosen before any run for sustained struggle and interviews. Both participants eventually asked for help. Struggle episodes were annotated as in [1], and afterwards each rewatched the recording and rated struggle and willingness to receive help. Each is replayed offline up to that verbal request.

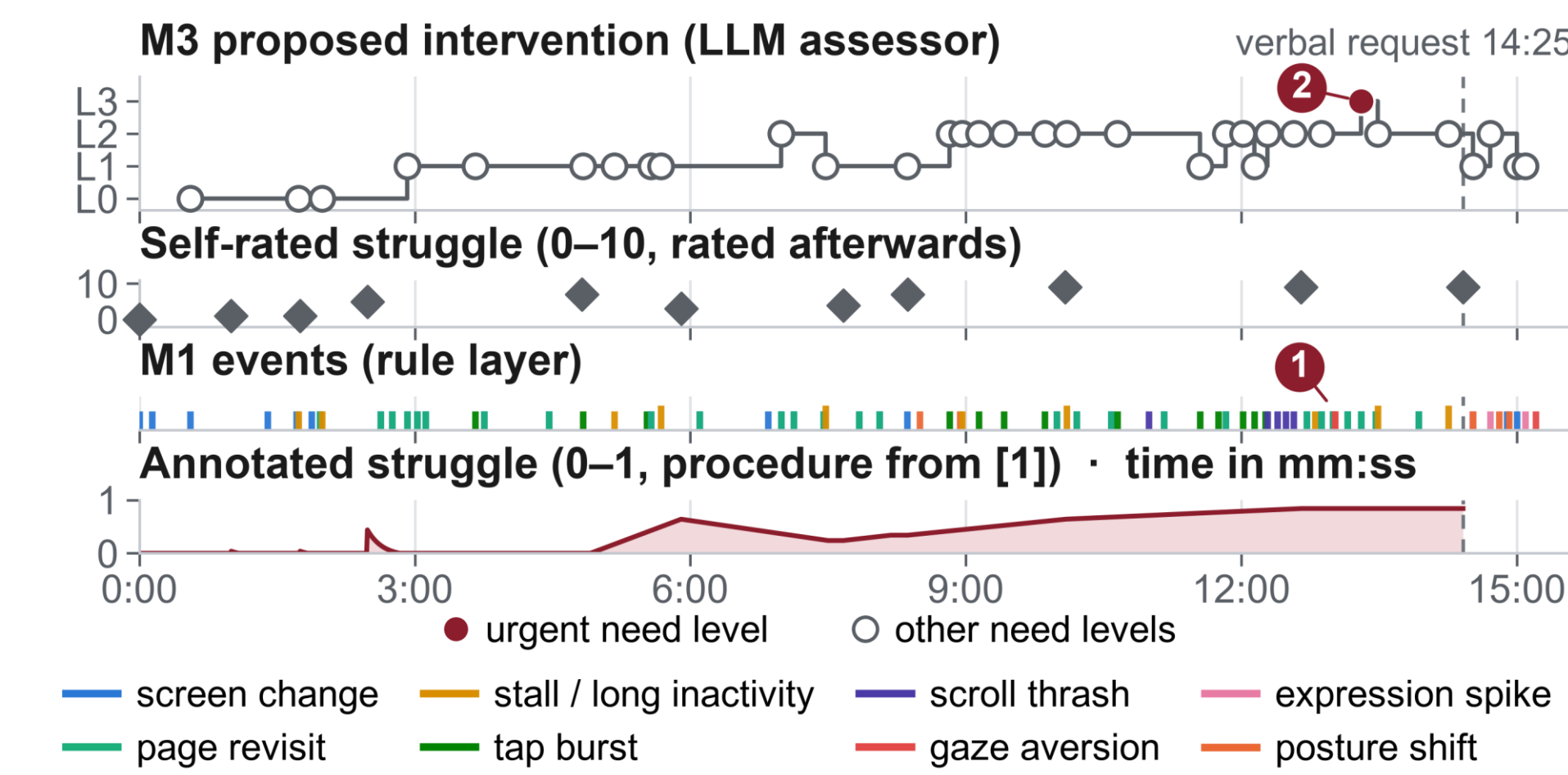
## 6 LESSONS FROM THE RUNS

- The first runs missed every tap burst. The rule counted clicks, and the kiosk reports touches. After the fix, M1 found 12 of Case A's 13 annotated struggle episodes and all 7 of Case B's, up from 1 and 3.
- On a non-case session, the assessor's full history inflated the need level. Weighting recent events restored the step down.
- Willingness to receive help peaked about 100 s (Case A) and 240 s (Case B) before the proposal reached L2.

### Case B, the long session (865 s)

- 13:01 narration: "turning toward the surroundings for help" citing E83 (posture shift) and E84 (gaze aversion)
- 13:18 the only L3 proposal in either case, 67 s before the request

Participant, afterwards: "... one more try, and if that failed, I'd call for help."



865 s to the verbal request (dashed line). The single L3 followed the 13:01 posture shift and gaze aversion, read as turning to the surroundings for help. After the request the proposal eased back to L1 as the screens moved forward.

**By the verbal request, narration, assessment, and the participants' own accounts agree.**